# Shahzaib Khan

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## Professional Summary

- Full-Stack AI Engineer with 3+ years of experience building production LLM and generative AI systems: AI agents, RAG pipelines, multi-agent workflows and voice AI, across hospitality, meetings, legal and AI governance.
- Currently Founding Senior Engineer at Bridge (AI meeting workspace, ∼500 users) and Lead Product Engineer at DreamDesk AI (voice and chat agents live at 3 hotels).
- Founding engineer at multiple AI startups and built AI products for a Y Combinator-funded startup. Strong in AI agent architecture, MCP integrations and end-to-end product delivery. Co-authored an arXiv paper on LLM-based test generation. Open to remote roles with US and UK overlap.

## Skills

**Programming Languages:** Python, JavaScript, TypeScript, C++, SQL

**Backend, Frontend & Cloud:** FastAPI, Node.js, React, Next.js, gRPC, Docker, AWS (S3, EC2, Bedrock), GCP (Cloud Run, Compute Engine), Azure, DigitalOcean

**Databases & Vector Databases:** PostgreSQL, MySQL, MongoDB, Supabase, Redis, Pinecone, pgvector, Weaviate

**Generative AI & LLMs:** AI agents, RAG, MCP, LangChain, LangGraph, LlamaIndex, Pydantic AI, fine-tuning, Voice AI (LiveKit, VAPI, Retell)

**Tools:** Git, GitHub Actions, Nginx, n8n, Twilio, Telnyx

## Work Experience

**Founding Senior Engineer**, Bridge – Remote, USA (Part-time) Aug 2025 – Present

- Joined as the first engineer and promoted to Founding Senior Engineer in Oct 2026. Leading engineering across product and growth for Bridge, an AI meeting workspace where teams and AI agents work together.
- Shipped the MVP in 2 months and scaled the platform from launch to ∼500 users. Optimized database queries, cutting response latency from 60s to under 2s (30x faster).
- Built LangChain agentic workflows that automate the full meeting lifecycle, from transcription (AssemblyAI, Deepgram) to summaries and follow-ups. Integrated MCP servers so agents take actions in ClickUp, Slack, Google Drive and Notion, plus third-party APIs like Apollo.
- Built the growth engineering stack: an automated content engine that researches, writes and publishes SEO blogs, and programmatic SEO/AEO/GEO pipelines (structured data, llms.txt, AI-crawler optimization) so Bridge ranks on Google and gets cited by ChatGPT, Perplexity and Google AI Overviews.
- Tech Stack: Node.js, Next.js, LangChain, MCP, AssemblyAI, Deepgram, PostgreSQL, DigitalOcean, GitHub Actions

**Lead Product Engineer**, DreamDesk AI – Remote, UK (Full-time) Jan 2026 – Present

- Led the engineering team and shipped the POC in 4 weeks. Designed the end-to-end product architecture and UX for a hospitality AI platform, scoping features and R&D directly with the product owner.
- Built multi-channel AI agents across phone calls, WhatsApp, email, web chat and Booking.com, live across 3 hometels of Second Nature (formerly room2), DreamDesk's parent company, in London, Belfast and Southampton.
- Voice agent responds in ∼1.7s end to end, including RAG retrieval and MCP tool calls. Integrated Mews & Guestline PMS for real-time reservation sync and built the RAG pipeline that ingests hotel knowledge bases.
- Tech Stack: FastAPI, Next.js, GCP (Cloud Run & Compute Engine), Pydantic AI, Pinecone, PostgreSQL, GitHub Actions

**Full-Stack AI Engineer**, AI Gov Sandbox – Remote, USA Apr 2025 – Nov 2025

- As the founding engineer, built a multi-agent system that automatically evaluates AI systems against the EU AI Act, US state-level AI regulations and global compliance frameworks.
- Integrated MIT AI Risk Repository, IBM AI Risk Atlas Nexus and Trusted AI's ART for security and risk evaluation. Dockerized each agent as an independent microservice.
- Tech Stack: Python, FastAPI, LangGraph, Docker, MongoDB, AWS (EC2, S3), GitHub Actions

**Full-Stack AI Engineer**, Foomotion LLC – Lahore, Pakistan Dec 2024 – Apr 2025

- Worked directly with Pearson Labs, a Y Combinator-funded legal AI startup: scoped requirements, led daily client syncs

and owned AI development end to end.

- Built Pearson Labs' chat-with-legal-documents RAG pipeline (LangChain, LlamaIndex, PyMuPDF) with Pinecone multi-tenancy, query expansion for better retrieval and tool calling, served through FastAPI and AWS S3 to a Next.js frontend.
- Built LLM-powered RAG and multi-agent systems on AWS Bedrock with LangGraph, plus LiveKit voice agents and n8n automation workflows.
- Tech Stack: Python, FastAPI, Next.js, LangGraph, LangChain, LlamaIndex, Pinecone, LiveKit, n8n, PostgreSQL, AWS (EC2, S3, Bedrock)

**AI Engineer**, Cyber Evangelists – Lahore, Pakistan Jun 2024 – Dec 2024

- Developed a real-time RAG voice calling agent on Node.js, integrating STT, TTS and voice cloning for automated customer conversations.
- Built the MVP of an Attack Surface Management platform (attackinsights.ai): FastAPI and gRPC microservices that run security tools independently, each Dockerized with PostgreSQL and Redis, plus a RAG agent (Llama 3.1) that queries live data from the tools.
- Tech Stack: Node.js, Python, FastAPI, gRPC, Docker, LangChain, PostgreSQL, Redis

**Full-Stack AI Engineer**, Freelance – Remote (Part-time, contract) Jun 2023 – Feb 2025

- Built MVPs for AI-powered products and an ERP system, spanning RAG, voice AI agents and multi-agent systems.
- Built a real-time voice calling agent for reves.ai on Node.js and WebSockets with STT, TTS, voice cloning and intent/emotion detection. Cut response latency and cost by 20%; deployed with Nginx on a VPS.
- Tech Stack: FastAPI, Node.js, Next.js, React, LangGraph, LangChain, LlamaIndex, VAPI, LiveKit, Deepgram, Play.ht, Twilio

**Software Engineering Intern**, DevDen – Faisalabad, Pakistan Jun 2023 – Aug 2023

- Built an admin dashboard for a POS system with React, featuring user management and sales analytics.

## Research

**Towards LLM-assisted High-Quality Property Generation for Solidity Smart Contracts** arXiv:2607.23308, Jul 2026

- Co-author (2nd of 6), with Dr. Hassan (DFKI Bremen, Germany) and Dr. Jalil (UAEU).
- Built an LLM pipeline that generates test invariants for property-based testing of Solidity smart contracts, with an agent that automatically resolves compilation errors in the output. Evaluated zero-shot, few-shot and prompt-chaining techniques and tuned hyperparameters to reduce hallucinations.
- Benchmarked against human-written properties with mutation testing: Gemini 1.5 Pro with prompt chaining reached a 25.99% mutation score vs 31.75% for human-written properties, and matched them on individual contracts (74.34% vs 74.83%).
- Tech Stack: Python, LangChain, LLMs, Azure ML, fuzzing, property-based testing, mutation testing

## Education

**National University of Computer and Emerging Sciences (FAST-NUCES)** Sep 2020 – Jun 2024

- BS in Computer Science

## Certifications

Data Analyst with Python (36-hour track) | Certificate DataCamp
LangChain Masterclass (Python) | Certificate Udemy

## Hackathons

**IBM watsonx Assistant Hackathon, lablab.ai** Nov 2024

- Built an HR bot that boosts employee engagement and streamlines HR by analysing satisfaction through daily feedback. Project